

COMPLETE STUDY GUIDE · VOL. 2

Intermediate College Mathematics

Functions · Trigonometry · Calculus · Differential Equations · Linear
Algebra · Statistics · 90+ Solved Problems

Prerequisites for this guide:

Comfortable with algebra (linear equations, quadratics, exponents).
Familiarity with school geometry and basic fractions. If not, start with
Vol. 1: Mathematics Fundamentals.

Functions & Pre-Calc

Trigonometry

Limits

Differentiation

Applications of Derivatives

Integration

Advanced Integration

Differential Equations

College Statistics

Linear Algebra

Video Links Included · Learn by Doing

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Prereq: Systems of Equations

Prerequisites: Algebra basics (linear equations, quadratics). Know what a variable is and how to solve for x .

Functions are the language of calculus. Before differentiating or integrating anything, you must be able to evaluate, compose, and invert functions, and understand exponential and logarithmic relationships.

Domain, Range & Function Notation

► Watch: Review of Functions

FUNCTION ESSENTIALS

Function Notation

$f(x)$ reads "f of x"; output for input x

Domain

All valid inputs (avoid $\div 0$ and $\sqrt{\text{(negative)}}$)

Range

All possible outputs

Composition

$$(f \circ g)(x) = f(g(x))$$

Inverse

Swap $x \leftrightarrow y$, solve for y : $f^{-1}(x)$

Verification

$$f(f^{-1}(x)) = x$$

1 Domain, composition and inverse

Given $f(x) = \sqrt{x-1}$ and $g(x) = x^2 + 3$, find: (a) Domain of f , (b) $(f \circ g)(x)$, (c) $f^{-1}(x)$.

STEP-BY-STEP SOLUTION

- a** Need $x - 1 \geq 0 \Rightarrow x \geq 1$. Domain: $[1, \infty)$
- b** $f(g(x)) = f(x^2 + 3) = \sqrt{(x^2 + 3) - 1} = \sqrt{x^2 + 2}$. Domain: all real x ($x^2 + 2 \geq 2 > 0$ always)
- c** Set $y = \sqrt{x-1}$. Square: $y^2 = x - 1 \Rightarrow x = y^2 + 1$. Swap: $f^{-1}(x) = x^2 + 1$, domain $x \geq 0$

Domain: $[1, \infty)$; $(f \circ g)(x) = \sqrt{(x^2 + 2)}$; $f^{-1}(x) = x^2 + 1$

2 Logarithm laws

Solve: $\log_3(2x + 1) = 3$ and simplify $\ln(e^4) - \ln(e)$.

STEP-BY-STEP SOLUTION

1 Convert log form: $3^3 = 2x + 1 \Rightarrow 27 = 2x + 1 \Rightarrow x = 13$

2 $\ln(e^4) - \ln(e) = 4 \ln e - \ln e = 4 - 1 = 3$

$x = 13$; $\ln(e^4) - \ln(e) = 3$

Module 1 — Formula Summary

Composition

$$(f \circ g)(x) = f(g(x))$$

Log $\leftrightarrow$ Exp

$$\log_b a = c \Leftrightarrow b^c = a$$

Log Product

$$\log(ab) = \log a + \log b$$

Log Quotient

$$\log(a/b) = \log a - \log b$$

Log Power

$$\log(a^n) = n \log a$$

Change of Base

$$\log_b a = \frac{\ln a}{\ln b}$$

Trigonometry (College Level)

Prerequisites: Basic SOHCAHTOA from school geometry, right triangle trigonometry, understanding of angles in degrees.

College trigonometry extends beyond right triangles. You'll master the unit circle (all quadrants), trig identities, inverse trig functions, and solving equations — all essential for calculus.

Unit Circle & Special Angles

► Watch: Basic Trigonometry

► Watch: Evaluating Trig Expressions

CORE TRIG FORMULAS

Pythagorean Identity

$$\sin^2 \theta + \cos^2 \theta = 1$$

Derived Identity 1

$$1 + \tan^2 \theta = \sec^2 \theta$$

Derived Identity 2

$$1 + \cot^2 \theta = \csc^2 \theta$$

Double Angle: sin

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

Double Angle: cos

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

Sum Formula: sin

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

Sum Formula: cos

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

Special: sin 30°

$$\frac{1}{2}, \cos 30^\circ = \frac{\sqrt{3}}{2}, \tan 30^\circ = \frac{1}{\sqrt{3}}$$

Special: sin 45°

$$\frac{\sqrt{2}}{2}, \cos 45^\circ = \frac{\sqrt{2}}{2}, \tan 45^\circ = 1$$

Special: sin 60°

$$\frac{\sqrt{3}}{2}, \cos 60^\circ = \frac{1}{2}, \tan 60^\circ = \sqrt{3}$$

1 Evaluating without a calculator

Evaluate exactly: (a) $\sin 150^\circ$ (b) $\cos 225^\circ$ (c) $\tan 300^\circ$

STEP-BY-STEP SOLUTION (Reference angle method)

a $150^\circ = 180^\circ - 30^\circ \rightarrow \text{Q2 (sin+)}. \text{Reference angle } 30^\circ. \sin 150^\circ = \sin 30^\circ = \frac{1}{2}$

- b** $225^\circ = 180^\circ + 45^\circ \rightarrow \text{Q3 (cos-). Reference angle } 45^\circ.$

$$\cos 225^\circ = -\cos 45^\circ = -\frac{\sqrt{2}}{2}$$

- c** $300^\circ = 360^\circ - 60^\circ \rightarrow \text{Q4 (tan-). Reference angle } 60^\circ.$

$$\tan 300^\circ = -\tan 60^\circ = -\sqrt{3}$$

$$\sin 150^\circ = 1/2, \cos 225^\circ = -\sqrt{2}/2, \tan 300^\circ = -\sqrt{3}$$

2 Proving a trig identity

Prove: $\frac{\sin \theta}{1 - \cos \theta} = \frac{1 + \cos \theta}{\sin \theta}$

STEP-BY-STEP SOLUTION

- 1** Cross-multiply (to verify): $\text{LHS} \times \sin \theta = \sin^2 \theta$; $\text{RHS} \times (1 - \cos \theta) = (1 + \cos \theta)(1 - \cos \theta) = 1 - \cos^2 \theta = \sin^2 \theta \checkmark$

- 2** Formal proof from LHS: Multiply by $\frac{1 + \cos \theta}{1 + \cos \theta}$:
- $$\frac{\sin \theta (1 + \cos \theta)}{1 - \cos^2 \theta} = \frac{\sin \theta (1 + \cos \theta)}{\sin^2 \theta} = \frac{1 + \cos \theta}{\sin \theta} = \text{RHS} \checkmark$$

Identity proved

Solving Trig Equations

► Watch: Trigonometric Identities

1 General solution

Solve $2 \sin^2 x - \sin x - 1 = 0$ for $0 \leq x < 2\pi$.

STEP-BY-STEP SOLUTION

- 1** Let $u = \sin x$: $2u^2 - u - 1 = 0 \Rightarrow (2u + 1)(u - 1) = 0$
- 2** $u = -\frac{1}{2}$ or $u = 1$
- 3** $\sin x = 1 \Rightarrow x = \frac{\pi}{2}$
- 4** $\sin x = -\frac{1}{2} \Rightarrow x = \frac{7\pi}{6}, \frac{11\pi}{6}$ (Q3 and Q4)

$$x = \pi/2, 7\pi/6, 11\pi/6$$

Module 2 — Formula Summary

Pythagorean

$$\sin^2 \theta + \cos^2 \theta = 1$$

Double Angle sin

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

Double Angle cos

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

Sum/Diff sin

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

Sum/Diff cos

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

ASTC Rule

Q1:All+, Q2:Sin+, Q3:Tan+, Q4:Cos+

Prerequisites: Functions and function notation (Module 1). Understanding of what "approaches" means intuitively.

Limits are the rigorous foundation of calculus. The derivative and the integral are both defined using limits. Understanding limits prevents common errors in calculus courses.

Evaluating Limits

► Watch: Limits – Video-Tutor.net

LIMIT LAWS & TECHNIQUES

Direct Substitution

If $f(c)$ is defined, $\lim_{x \rightarrow c} f(x) = f(c)$

Factor & Cancel

For $\frac{0}{0}$: factor numerator/denominator, cancel common factor

L'Hôpital's Rule

If $\frac{0}{0}$ or $\frac{\infty}{\infty}$: $\lim_{x \rightarrow c} \frac{f}{g} = \lim_{x \rightarrow c} \frac{f'}{g'}$

Squeeze Theorem

If $g(x) \leq f(x) \leq h(x)$ and $\lim g = \lim h = L$, then $\lim f = L$

Important Limit 1

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

Important Limit 2

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$$

1 Factor and cancel (0/0 form)

Evaluate: $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

STEP-BY-STEP SOLUTION

- 1 Direct sub: $\frac{9-9}{3-3} = \frac{0}{0} \rightarrow$ Indeterminate. Factor numerator.
- 2 $\frac{(x+3)(x-3)}{x-3} = x+3$ (for $x \neq 3$)
- 3 $\lim_{x \rightarrow 3} (x+3) = 6$

Answer: 6

2 L'Hôpital's Rule

Evaluate: $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$

STEP-BY-STEP SOLUTION

- 1 Direct sub gives $\frac{0}{0}$. Apply L'Hôpital: $\lim_{x \rightarrow 0} \frac{e^x - 1}{2x}$
- 2 Still $\frac{0}{0}$. Apply again: $\lim_{x \rightarrow 0} \frac{e^x}{2} = \frac{e^0}{2} = \frac{1}{2}$

Answer: 1/2

3 Continuity check

Is $f(x) = \begin{cases} x^2 - 1 & x < 2 \\ 3 & x = 2 \\ 2x - 1 & x > 2 \end{cases}$ continuous at $x = 2$?

STEP-BY-STEP SOLUTION (3 conditions for continuity)

- 1 $f(2) = 3 \checkmark$ (defined)
- 2 Left limit: $\lim_{x \rightarrow 2^-} (x^2 - 1) = 3$. Right limit: $\lim_{x \rightarrow 2^+} (2x - 1) = 3$. So $\lim_{x \rightarrow 2} f(x) = 3 \checkmark$
- 3 $\lim_{x \rightarrow 2} f(x) = 3 = f(2) \checkmark$ All conditions met.

f is continuous at x = 2

Module 3 — Formula Summary

L'Hôpital's Rule

$$\lim \frac{f}{g} \stackrel{0/0}{=} \lim \frac{f'}{g'}$$

sin limit

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

Continuity

Defined + Limit exists + Equal

Sandwich Theorem

$g \leq f \leq h$ and $\lim g = \lim h = L \Rightarrow \lim f = L$

Prerequisites: Limits (Module 3), function notation (Module 1), trig identities (Module 2).

The derivative measures instantaneous rate of change. It is the most-used operation in applied mathematics, physics, and engineering. You must master all differentiation rules.

Differentiation Rules

► Watch: Derivatives – Video-Tutor.net

DIFFERENTIATION RULES

Definition

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Power Rule

$$\frac{d}{dx} x^n = nx^{n-1}$$

Constant

$$\frac{d}{dx} c = 0$$

Product Rule

$$(uv)' = u'v + uv'$$

Quotient Rule

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

Chain Rule

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$$

$\frac{d}{dx} \sin x$

$$\cos x$$

$\frac{d}{dx} \cos x$

$$-\sin x$$

$\frac{d}{dx} \tan x$

$$\sec^2 x$$

$\frac{d}{dx} e^x$

$$e^x$$

$\frac{d}{dx} \ln x$

$$\frac{1}{x}$$

$\frac{d}{dx} a^x$

$$a^x \ln a$$

1 Product rule + chain rule

Differentiate: $y = x^3 \cdot e^{2x}$

STEP-BY-STEP SOLUTION

- 1 Product rule: $u = x^3$, $u' = 3x^2$; $v = e^{2x}$, $v' = 2e^{2x}$ (chain rule)
- 2 $y' = u'v + uv' = 3x^2e^{2x} + x^3 \cdot 2e^{2x} = e^{2x}(3x^2 + 2x^3)$
- 3 Factor: $y' = x^2e^{2x}(3 + 2x)$

$$y' = x^2e^{2x}(3 + 2x)$$

2 Quotient rule

Find $\frac{d}{dx} \left[\frac{\sin x}{x^2 + 1} \right]$

STEP-BY-STEP SOLUTION

- 1 $u = \sin x$, $u' = \cos x$; $v = x^2 + 1$, $v' = 2x$
- 2 $\frac{d}{dx} = \frac{\cos x(x^2 + 1) - \sin x \cdot 2x}{(x^2 + 1)^2} = \frac{(x^2 + 1) \cos x - 2x \sin x}{(x^2 + 1)^2}$

$$\frac{(x^2 + 1) \cos x - 2x \sin x}{(x^2 + 1)^2}$$

3 Implicit differentiation

Given $x^2 + y^2 = 25$, find $\frac{dy}{dx}$ at the point (3, 4).

STEP-BY-STEP SOLUTION

- 1 Differentiate both sides w.r.t. x : $2x + 2y \frac{dy}{dx} = 0$
- 2 $\frac{dy}{dx} = -\frac{x}{y}$. At (3,4): $\frac{dy}{dx} = -\frac{3}{4}$

$$dy/dx = -3/4$$

Module 4 — Formula Summary

Power Rule

Product Rule

$$(uv)' = u'v + uv'$$

$$\frac{d}{dx} x^n = nx^{n-1}$$

Quotient Rule

$$(u/v)' = (u'v - uv')/v^2$$

Chain Rule

$$\frac{d}{dx} f(g) = f'(g) \cdot g'$$

d/dx e^x

$$e^x$$

d/dx ln x

$$1/x$$

d/dx sin

$$\cos x$$

d/dx cos

$$-\sin x$$

Applications of Derivatives

Prerequisites: Differentiation (Module 4). Ability to find and classify critical points.

Derivatives tell us where functions increase, decrease, and turn around. This is the heart of optimization — finding maximum profit, minimum cost, fastest speed — used in every engineering and business application.

Optimization & Critical Points

► Watch: Applications of Derivatives

KEY TESTS

Critical Points

Set $f'(x) = 0$ or $f'(x)$ undefined

1st Derivative Test

$+\rightarrow-$ at CP: local max; $-\rightarrow+$ at CP: local min

2nd Derivative Test

$f''(c) < 0$: local max; $f''(c) > 0$: local min

Concave Up

$f''(x) > 0$

Concave Down

$f''(x) < 0$

Inflection Point

$f''(x) = 0$ and concavity changes

Mean Value Theorem

$f'(c) = \frac{f(b) - f(a)}{b - a}$ for some $c \in (a, b)$

1 Find maxima and minima

Find all local extrema of $f(x) = x^3 - 6x^2 + 9x + 2$ and classify them.

STEP-BY-STEP SOLUTION

1 $f'(x) = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = 3(x - 1)(x - 3)$

2 Critical points: $x = 1$ and $x = 3$

3 $f''(x) = 6x - 12$. At $x = 1$: $f''(1) = -6 < 0 \rightarrow$ Local Maximum. At $x = 3$: $f''(3) = 6 > 0 \rightarrow$ Local Minimum.

4 $f(1) = 1 - 6 + 9 + 2 = 6$. $f(3) = 27 - 54 + 27 + 2 = 2$

Local Max at (1, 6); Local Min at (3, 2)

2 Real-world optimization

A farmer has 200 m of fencing to enclose a rectangular field. One side borders a river (no fence needed there). Find the dimensions that maximize the area.

STEP-BY-STEP SOLUTION

1 Let width = x , length = y . Constraint: $2x + y = 200 \Rightarrow y = 200 - 2x$

2 Area: $A = xy = x(200 - 2x) = 200x - 2x^2$

3 $A' = 200 - 4x = 0 \Rightarrow x = 50$. $A'' = -4 < 0 \rightarrow$ Maximum.

4 $y = 200 - 100 = 100$. Area = $50 \times 100 = 5000 \text{ m}^2$

Width = 50 m, Length = 100 m, Max Area = 5000 m²

Module 5 — Formula Summary

Critical Points

$$f'(x) = 0 \text{ or undefined}$$

Local Max (2nd test)

$$f''(c) < 0$$

Local Min (2nd test)

$$f''(c) > 0$$

MVT

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Integration (Fundamentals)

Prerequisites: Differentiation (Module 4). Antiderivatives are the reverse of derivatives.

Integration computes area under curves, total accumulated change, and is the inverse of differentiation. The Fundamental Theorem of Calculus (FTC) connects these two operations beautifully.

Antiderivatives & Definite Integrals

► Watch: Integration – Video-Tutor.net

INTEGRATION RULES

Power Rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$\int 1/x dx$

$$\ln |x| + C$$

$\int e^x dx$

$$e^x + C$$

$\int \sin x dx$

$$-\cos x + C$$

$\int \cos x dx$

$$\sin x + C$$

$\int \sec^2 x dx$

$$\tan x + C$$

FTC Part 1

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

FTC Part 2

$$\int_a^b f(x) dx = F(b) - F(a)$$

u-Substitution

Let $u = g(x)$, then

$$\int f(g(x))g'(x) dx = \int f(u) du$$

1 Indefinite integral with u-substitution

Evaluate: $\int 2x \cos(x^2) dx$

STEP-BY-STEP SOLUTION

1 Let $u = x^2 \Rightarrow du = 2x dx$. Substitute.

2 $\int \cos(u) du = \sin(u) + C = \sin(x^2) + C$

$$\sin(x^2) + C$$

2 Definite integral (FTC)

Evaluate: $\int_1^4 \left(3x^2 - \frac{2}{x} \right) dx$

STEP-BY-STEP SOLUTION

1 Antiderivative: $F(x) = x^3 - 2 \ln |x|$

2 $F(4) - F(1) = (64 - 2 \ln 4) - (1 - 0) = 63 - 2 \ln 4$

3 $\approx 63 - 2(1.386) \approx 63 - 2.773 \approx 60.23$

$$63 - 2 \ln 4 \approx 60.23$$

3 Area between curves

Find the area between $y = x^2$ and $y = x + 2$.

STEP-BY-STEP SOLUTION

1 Intersections: $x^2 = x + 2 \Rightarrow x^2 - x - 2 = 0 \Rightarrow (x - 2)(x + 1) = 0$. So $x = -1$ and $x = 2$.

2 Upper - Lower: $\int_{-1}^2 [(x + 2) - x^2] dx$

3 $= \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2 = \left(2 + 4 - \frac{8}{3} \right) - \left(\frac{1}{2} - 2 + \frac{1}{3} \right) = \frac{10}{3} - \left(-\frac{7}{6} \right) = \frac{9}{2}$

$$\text{Area} = 9/2 \text{ square units}$$

Module 6 — Formula Summary

Power Rule $\int$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$\int e^x$

$$e^x + C$$

$\int \sin x$

$$-\cos x + C$$

$\int \cos x$

$$\sin x + C$$

FTC

$$\int_a^b f dx = F(b) - F(a)$$

u-sub

Let $u = g(x)$, replace $g'(x)dx = du$

Advanced Integration Techniques

Prerequisites: Basic integration (Module 6), trig identities (Module 2), partial fractions require factoring skills.

When u-substitution isn't enough, these three powerful techniques handle integrals of products, rational functions, and expressions with square roots of quadratics.

Integration by Parts

► Watch: Integration Techniques

INTEGRATION BY PARTS

Formula

$$\int u \, dv = uv - \int v \, du$$

LIATE Rule (choose u)

Logarithmic → Inverse trig →
Algebraic → Trig → Exponential

Partial Fractions

Decompose $\frac{P(x)}{Q(x)}$ into simpler fractions; integrate each

Trig Sub: $\sqrt{a^2 - x^2}$

Let $x = a \sin \theta$

Trig Sub: $\sqrt{a^2 + x^2}$

Let $x = a \tan \theta$

Trig Sub: $\sqrt{x^2 - a^2}$

Let $x = a \sec \theta$

1 Integration by parts

Evaluate: $\int x \cdot e^x \, dx$

STEP-BY-STEP SOLUTION

① LIATE: Algebraic before Exponential. Let $u = x \Rightarrow du = dx$;
 $dv = e^x dx \Rightarrow v = e^x$

② $\int x e^x dx = x e^x - \int e^x dx = x e^x - e^x + C = e^x(x - 1) + C$

$$e^x(x - 1) + C$$

Evaluate: $\int \frac{3x + 5}{(x + 1)(x + 2)} dx$

STEP-BY-STEP SOLUTION

1 Decompose: $\frac{3x + 5}{(x + 1)(x + 2)} = \frac{A}{x + 1} + \frac{B}{x + 2}$

2 $3x + 5 = A(x + 2) + B(x + 1)$. At $x = -1$: $2 = A(1) \Rightarrow A = 2$. At $x = -2$: $-1 = B(-1) \Rightarrow B = 1$

3 $\int \left[\frac{2}{x + 1} + \frac{1}{x + 2} \right] dx = 2 \ln |x + 1| + \ln |x + 2| + C$

$$2 \ln |x + 1| + \ln |x + 2| + C$$

Module 7 — Formula Summary

By Parts

$$\int u dv = uv - \int v du$$

LIATE

Log $\rightarrow$ InvTrig $\rightarrow$ Alg $\rightarrow$ Trig $\rightarrow$ Exp

Trig Sub $\sqrt{a^2 - x^2}$

$$x = a \sin \theta$$

Trig Sub $\sqrt{a^2 + x^2}$

$$x = a \tan \theta$$

Differential Equations

Prerequisites: Differentiation (Module 4) and integration (Module 6). Understanding of exponential and logarithmic functions.

A differential equation relates a function to its derivatives. They model population growth, cooling, electrical circuits, and mechanical vibrations. These appear in every engineering discipline.

Separable & Linear First-Order ODEs

► Watch: Differential Equations

ODE METHODS

Separable ODE

$$\frac{dy}{dx} = f(x)g(y) \Rightarrow \int \frac{dy}{g(y)} = \int f(x) dx$$

Linear 1st Order

$$\frac{dy}{dx} + P(x)y = Q(x)$$

Integrating Factor

$$\mu(x) = e^{\int P(x)dx}$$

Solution

$$y = \frac{1}{\mu} \int \mu \cdot Q dx$$

Exponential Model

$$\frac{dy}{dt} = ky \Rightarrow y = Ce^{kt}$$

Newton's Cooling

$$\frac{dT}{dt} = k(T - T_{\text{env}})$$

1 Separable ODE

Solve: $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 3$.

STEP-BY-STEP SOLUTION

- 1 Separate: $y dy = x dx$
- 2 Integrate both sides: $\frac{y^2}{2} = \frac{x^2}{2} + C \rightarrow y^2 = x^2 + K$
- 3 Apply IC $y(0) = 3$: $9 = 0 + K \Rightarrow K = 9$
- 4 $y^2 = x^2 + 9 \Rightarrow y = \sqrt{x^2 + 9}$ (positive since $y(0)=3>0$)

$$y = \sqrt{x^2 + 9}$$

2 Exponential growth / decay

A bacteria culture starts with 500 cells and doubles every 3 hours. How many cells after 9 hours?

STEP-BY-STEP SOLUTION

- 1 Model: $N = N_0 \cdot 2^{t/3}$. Initial $N_0 = 500$.
- 2 At $t = 9$: $N = 500 \cdot 2^{9/3} = 500 \cdot 2^3 = 500 \times 8 = 4000$

4,000 cells after 9 hours

3 Linear first-order ODE

Solve: $\frac{dy}{dx} + \frac{2}{x}y = x^2$

STEP-BY-STEP SOLUTION

- 1 Integrating factor: $\mu = e^{\int \frac{2}{x} dx} = e^{2 \ln x} = x^2$
- 2 Multiply: $\frac{d}{dx}(x^2 y) = x^2 \cdot x^2 = x^4$
- 3 Integrate: $x^2 y = \frac{x^5}{5} + C \Rightarrow y = \frac{x^3}{5} + \frac{C}{x^2}$

$$y = \frac{x^3}{5} + \frac{C}{x^2}$$

Module 8 — Formula Summary

Separable

$$\int \frac{dy}{g(y)} = \int f(x) dx$$

Integrating Factor

$$\mu = e^{\int P dx}$$

Linear ODE Solution

$$y = \frac{1}{\mu} \int \mu Q dx$$

Exponential Growth

$$y = Ce^{kt}$$

Prerequisites: Basic mean/median/mode from school statistics. Understanding of probability fundamentals.

College statistics extends into probability distributions, hypothesis testing, and confidence intervals — the statistical tools used in data science, research papers, and quality control.

Probability Distributions

► Watch: Discrete Probability Distributions

► Watch: Normal Distributions

DISTRIBUTIONS

Binomial $P(X=k)$

$$\binom{n}{k} p^k (1-p)^{n-k}$$

Binomial Mean

$$\mu = np$$

Binomial Variance

$$\sigma^2 = np(1-p)$$

Poisson $P(X=k)$

$$\frac{e^{-\lambda} \lambda^k}{k!}$$

Z-Score

$$Z = \frac{X - \mu}{\sigma}$$

Standard Error

$$SE = \frac{\sigma}{\sqrt{n}}$$

95% Conf. Interval

$$\bar{x} \pm 1.96 \cdot \frac{\sigma}{\sqrt{n}}$$

t-interval

$$\bar{x} \pm t^* \cdot \frac{s}{\sqrt{n}}$$

1 Binomial distribution

A fair coin is tossed 10 times. Find: (a) $P(\text{exactly 6 heads})$, (b) Mean and standard deviation.

STEP-BY-STEP SOLUTION

a $n = 10, p = 0.5, k = 6.$

$$P(X = 6) = \binom{10}{6} (0.5)^6 (0.5)^4 = 210 \times (0.5)^{10} = \frac{210}{1024} \approx 0.205$$

b Mean: $\mu = 10 \times 0.5 = 5$. SD: $\sigma = \sqrt{10 \times 0.5 \times 0.5} = \sqrt{2.5} \approx 1.58$

$P(X=6) \approx 0.205$; Mean=5; SD ≈ 1.58

2 Z-score and hypothesis testing

A factory claims its light bulbs last 1000 hours ($\sigma=50$). A sample of 25 bulbs gives $\bar{x} = 980$ hrs. Test at $\alpha=0.05$ whether the true mean is less than 1000.

STEP-BY-STEP SOLUTION

1 $H_0: \mu=1000$; $H_1: \mu<1000$ (left-tailed)

$$2 \quad Z = \frac{980 - 1000}{50/\sqrt{25}} = \frac{-20}{10} = -2.0$$

3 Critical value at $\alpha=0.05$ (left tail): $Z_{crit} = -1.645$

4 Since $-2.0 < -1.645$, we REJECT H_0 . Sufficient evidence that mean<1000.

Reject H_0 — evidence that mean lifetime < 1000 hrs

Confidence Intervals

► Watch: Confidence Intervals

1 95% Confidence interval

A sample of 36 students has mean exam score $\bar{x} = 72$ with $\sigma=12$. Construct a 95% CI for the true mean.

STEP-BY-STEP SOLUTION

$$1 \quad SE = \frac{12}{\sqrt{36}} = \frac{12}{6} = 2$$

$$2 \quad 95\% \text{ CI: } 72 \pm 1.96 \times 2 = 72 \pm 3.92$$

$$3 \quad \text{Interval: } (68.08, 75.92)$$

95% CI: (68.08, 75.92)

Module 9 — Formula Summary

Binomial P(X=k)

$$\binom{n}{k} p^k (1-p)^{n-k}$$

Binomial Mean/Var

$$\mu = np, \sigma^2 = np(1-p)$$

Z-Score

$$Z = (X - \mu) / \sigma$$

Standard Error

$$SE = \sigma / \sqrt{n}$$

95% CI

$$\bar{x} \pm 1.96 \cdot \sigma / \sqrt{n}$$

Poisson

$$P(X = k) = e^{-\lambda} \lambda^k / k!$$

Prerequisites: Systems of equations (school algebra). Basic arithmetic with fractions and negative numbers.

Linear algebra is the mathematics of vectors and matrices. It powers AI/ML (neural networks), computer graphics, engineering simulations, cryptography, and data compression.

Matrices & Determinants

MATRIX OPERATIONS

Matrix Multiply

$$(AB)_{ij} = \sum_k A_{ik} B_{kj}$$

2×2 Determinant

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

3×3 Determinant

Cofactor expansion along any row/column

2×2 Inverse

$$A^{-1} = \frac{1}{\det A} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Cramer's Rule

$$x_i = \frac{\det(A_i)}{\det(A)}$$

Eigenvalue

$$\det(A - \lambda I) = 0$$

Eigenvector

$$(A - \lambda I)\mathbf{v} = \mathbf{0}$$

1 Matrix multiplication & determinant

Given $A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix}$, find AB and $\det(A)$.

STEP-BY-STEP SOLUTION

$$\text{① } AB = \begin{pmatrix} 2(1) + 3(2) & 2(-1) + 3(0) \\ 1(1) + 4(2) & 1(-1) + 4(0) \end{pmatrix} = \begin{pmatrix} 8 & -2 \\ 9 & -1 \end{pmatrix}$$

$$\text{② } \det(A) = 2 \times 4 - 3 \times 1 = 8 - 3 = 5$$

$$AB = \begin{pmatrix} 8 & -2 \\ 9 & -1 \end{pmatrix}, \det(A)=5$$

2 Eigenvalues and eigenvectors

Find the eigenvalues of $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ and the eigenvector for each.

STEP-BY-STEP SOLUTION

- 1 Characteristic equation: $\det(A - \lambda I) = 0$
- 2 $\det \begin{pmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{pmatrix} = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2) = 0$
- 3 Eigenvalues: $\lambda_1 = 5, \lambda_2 = 2$
- 4 For $\lambda_1 = 5$: $(A - 5I)\mathbf{v} = 0 \Rightarrow \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \mathbf{v} = 0 \Rightarrow v_1 = v_2$. Eigenvector:
 $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$
- 5 For $\lambda_2 = 2$: $(A - 2I)\mathbf{v} = 0 \Rightarrow \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \mathbf{v} = 0 \Rightarrow 2v_1 + v_2 = 0$. Eigenvector:
 $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$

$\lambda_1=5$ with eigenvector $[1,1]^T$; $\lambda_2=2$ with eigenvector $[1,-2]^T$

3 Gaussian elimination

Solve the system using Gaussian elimination:

$$x + y + z = 6, \quad 2x - y + z = 3, \quad x + 2y - z = 2$$

STEP-BY-STEP SOLUTION

- 1 Augmented matrix: $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 3 \\ 1 & 2 & -1 & 2 \end{array} \right)$
- 2 $R_2 \leftarrow R_2 - 2R_1; R_3 \leftarrow R_3 - R_1$: $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & -1 & -9 \\ 0 & 1 & -2 & -4 \end{array} \right)$
- 3 $R_3 \leftarrow R_3 + R_2/3$: $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & -1 & -9 \\ 0 & 0 & -7/3 & -7 \end{array} \right)$
- 4 Back-substitute: $z = 3, -3y - 3 = -9 \Rightarrow y = 2, x + 2 + 3 = 6 \Rightarrow x = 1$

$$x = 1, y = 2, z = 3$$

Module 10 — Formula Summary

2×2 Det

$$ad - bc$$

2×2 Inverse

$$\frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Eigenvalue Eq

$$\det(A - \lambda I) = 0$$

Eigenvector

$$(A - \lambda I)\mathbf{v} = \mathbf{0}$$

Cramer's Rule

$$x_i = \det(A_i) / \det(A)$$

Rank

Number of non-zero rows in row-echelon form

Master Formula Sheet — College Mathematics

Functions & Pre-Calculus

Composition

$$(f \circ g)(x) = f(g(x))$$

$\log \Leftrightarrow \exp$

$$\log_b a = c \Leftrightarrow b^c = a$$

Log Product

$$\log(ab) = \log a + \log b$$

Change of Base

$$\log_b a = \frac{\ln a}{\ln b}$$

Trigonometry

Pythagorean

$$\sin^2 \theta + \cos^2 \theta = 1$$

$1 + \tan^2$

$$1 + \tan^2 \theta = \sec^2 \theta$$

Double sin

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

Double cos

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

Sum sin

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

Sum cos

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

Limits

L'Hôpital

$$\lim(f/g) \stackrel{0/0}{=} \lim(f'/g')$$

$\sin x / x$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

Continuity

$f(c)$ defined, limit exists, equal

Differentiation

Power

$$\frac{d}{dx} x^n = nx^{n-1}$$

Product

$$(uv)' = u'v + uv'$$

Quotient

$$(u/v)' = (u'v - uv')/v^2$$

Chain

$$\frac{d}{dx} f(g) = f'(g) \cdot g'$$

d/dx e^x

$$e^x$$

d/dx ln x

$$1/x$$

d/dx sin

$$\cos x$$

d/dx cos

$$-\sin x$$

Integration

Power

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

FTC

$$\int_a^b f = F(b) - F(a)$$

By Parts

$$\int u dv = uv - \int v du$$

∫ sin x

$$-\cos x + C$$

∫ cos x

$$\sin x + C$$

∫ e^x

$$e^x + C$$

∫ 1/x

$$\ln |x| + C$$

Differential Equations & Statistics

Exp Growth

$$y = Ce^{kt}$$

Integrating Factor

$$\mu = e^{\int P dx}$$

Z-Score

$$Z = (X - \mu)/\sigma$$

95% CI

$$\bar{x} \pm 1.96 \cdot \sigma/\sqrt{n}$$

Binomial Mean

$$\mu = np$$

Eigenvalue Eq

$$\det(A - \lambda I) = 0$$

2×2 Inverse

$$\frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

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For foundational topics, see Vol. 1: Mathematics Fundamentals